Unit: 6 Unit 6: Systems of Linear Equations and Inequalities **Name:** ________________
Topic: Solving Systems Using Elimination and Substitution **Date:** __________

Homework (Carolina Reaper)

Q1. Solve the system: $x + 2y = 7$, $3x - 2y = 5$. What is x?
(A) 1
(B) 2
(C) 3
(D) 4
(E) 5

Q2. A music festival has two stages. Stage 1 has x attendees and Stage 2 has y attendees, where $x + y = 1000$ and $2x - y = 600$. What is y?
(A) 200
(B) 250
(C) 300
(D) 400
(E) 500

Q3. Solve the system using elimination: $2x + 3y = 12$, $x - 2y = -3$. What is y?
(A) 1
(B) 2
(C) 3
(D) 4
(E) 5

Q4. A painter has x hours to paint a mural and y hours to paint a canvas, where $x + 2y = 10$ and $x - y = 2$. What is x?
(A) 2
(B) 4
(C) 6
(D) 8
(E) 10

Q5. The rhythm pattern of a song has x beats in the first measure and y beats in the second measure, where $x + y = 16$ and $2x - 3y = -4$. What is y?
(A) 4
(B) 6
(C) 8
(D) 10
(E) 12

Q6. Solve the system: $x - 2y = 3$, $3x + 2y = 15$. What is x?
(A) 3
(B) 5
(C) 6
(D) 9
(E) 12

Q7. Determine the number of solutions to the system: $x+y = 5$, $2x + 2y = 8$. What is the number of solutions?
(A) 0
(B) 1
(C) Infinitely many
(D) No solution
(E) Two solutions

Q8. Solve the system using substitution: $x - 3y = 2$, $2x + 2y = 12$. What is y?
(A) 1
(B) 2
(C) 3
(D) 4
(E) 5

Open Ended Questions (FRQ)

Q9. A frequency analyzer has two frequencies, x and y, where $x + 2y = 20$ and $3x - y = 15$. Solve for x and y and determine the number of solutions to the system.

Q10. A canvas has a width of x inches and a height of y inches, where $2x + 3y = 50$ and $x - 2y = -10$. Solve for x and y using elimination and determine the area of the canvas.

Chef's Tips

- Tip 1: Check for infinitely many solutions or no solution.
- Tip 2: Use substitution or elimination to solve the system.
- Tip 3: Verify the solution by plugging it back into the original equations.